

ANI

NAGASTRA-1

Kamikaze drones made by Solar Industries. Army procured 480 units

- **COMBAT RADIUS:** 30 km
- **WEAPONS:** High explosive fragmenting warhead
- **TACTICAL USE:** Used for short-range precision strikes

NETRA

Tactical ISR drone made by IdeaForge

- **COMBAT RADIUS:** 2.5 km
- **TACTICAL USE:** Used for counter-insurgency operations



infantry, 30 'Bhairav' light commando battalions, each roughly 250-strong and trained in operating drones, are being formed to provide lightning fast strike options. 'Rudra' brigades are another innovation—all-arms formations combining elements of mechanised infantry, special forces, artillery and armed with UAVs ideal for multi-domain operations. Similarly, artillery regiments would have 'Divyastra' batteries—pairing kamikaze drones with long-range guns—intended to provide deep-strike capability. Lastly, 'Shaktibaan' formations are pure drone/ loitering munitions strike units.

Reconnaissance platoons will now have surveillance and strike drones and the Corps of Electronics and Mechanical Engineers will also get drone sections for field-level repairs and terrain mapping. And utility drones would resupply forward elements.

However, not all news about India's drone force is positive. Operation Sindoor exposed significant shortcomings in key areas. Cybersecurity was a major concern. Drones are open to cyber threats like denial of service, as well as phishing and spoofing (installation of malware to procure sensitive information). Many indigenous drones were rendered inactive due to Pakistani GPS jamming, which manipulated their navigation systems by emitting false signals. Significantly, the Israeli drones were impervious to this.

Another limitation was the inability to operate drones in swarms—a key factor in combat. Each drone operated as an isolated entity, communicating only with its operator. This lack of interoperability severely restricted their effectiveness.

At a recent seminar in Pune, defence secretary R.K. Singh highlighted capability gaps emerging out of Op. Sindoor. "These include electronic warfare, counter-unmanned systems and a stronger manufacturing base for military-grade drones that can survive in GPS-denied and contested environments, and low-level radars," he said.

The army's grand drone plans could amount to little if these challenges are not overcome.

Ailing Indian Drones

India has 840 drone companies, but it is hobbled by deep structural flaws. Most Indian manufacturers are integrators, and depend on imported components like motors, sensors, flight controllers, batteries and rare-earth magnets—comprising 60-70 per cent of critical components and mostly sourced from China. This reliance on Chinese supply chains presents a strategic vulnerability—it weakens preparedness and results in a shortfall of critical spares. "Capabilities of foreign weapons are known to all, and adversaries can predict our tactics.... If we develop our own systems...that may add an element of surprise," Chief of Defence Staff Gen. Anil Chauhan said at a recent seminar.

The absence of domestic production of advanced electronics keeps costs high, while a rigid certification regime for sub-systems (like flight controllers and GPS), and policy uncertainty around BVLOS (Beyond Visual Line of Sight) operations of drones—essential for operations in enemy territory—delay innovation. With small-scale production and imported inputs, Indian drones often end up costlier yet less reliable than foreign models.

"Funding for deep-tech R&D is scarce, and a shortage of skilled



"The military must first articulate precise requirements, specifying types of indigenous components they seek in Indian drones. They should allow the industry time to develop technologies"

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